

# CAUDAL ANESTHESIA

Adugna A.  
Assi.Professor of Anesthesia  
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# Introduction

- Caudal anesthesia has been used for many years and is the easiest and safest approach to the epidural space
- When correctly performed there is little danger of either the spinal cord or dura being damaged
- It is used to provide peri and post operative analgesia in adults and most commonly in children
- It can be used as a sole anesthetic for some procedures, or it may be combined with general anesthesia.

# Indications

- **Lower abdominal surgeries**
- **Pediatric urethral and perineal area surgeries**
- **Lower extremity surgery**

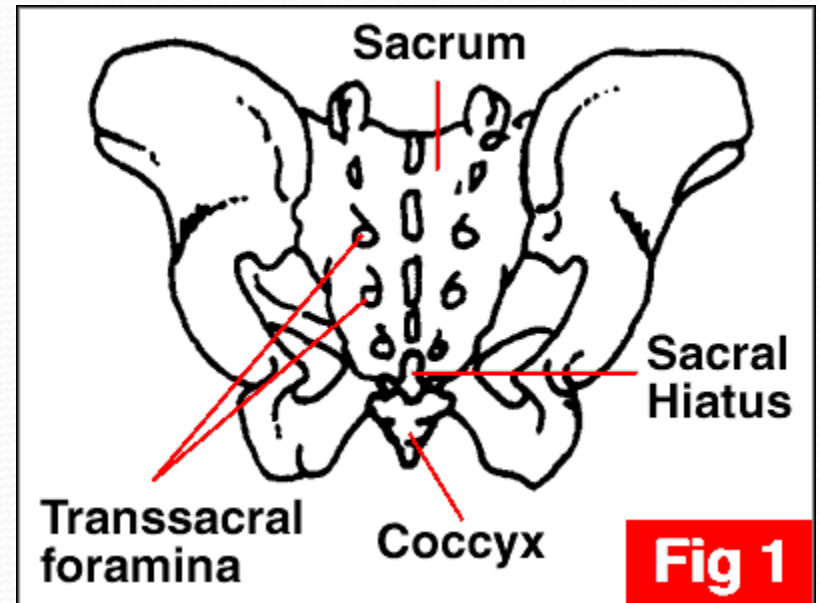


# Contraindications

- Infection near the site of the needle insertion.
- Coagulopathy or anti coagulation.
- Pilonidal cyst
- Congenital abnormalities of the lower spine or meninges, because of the unclear or impalpable anatomy.

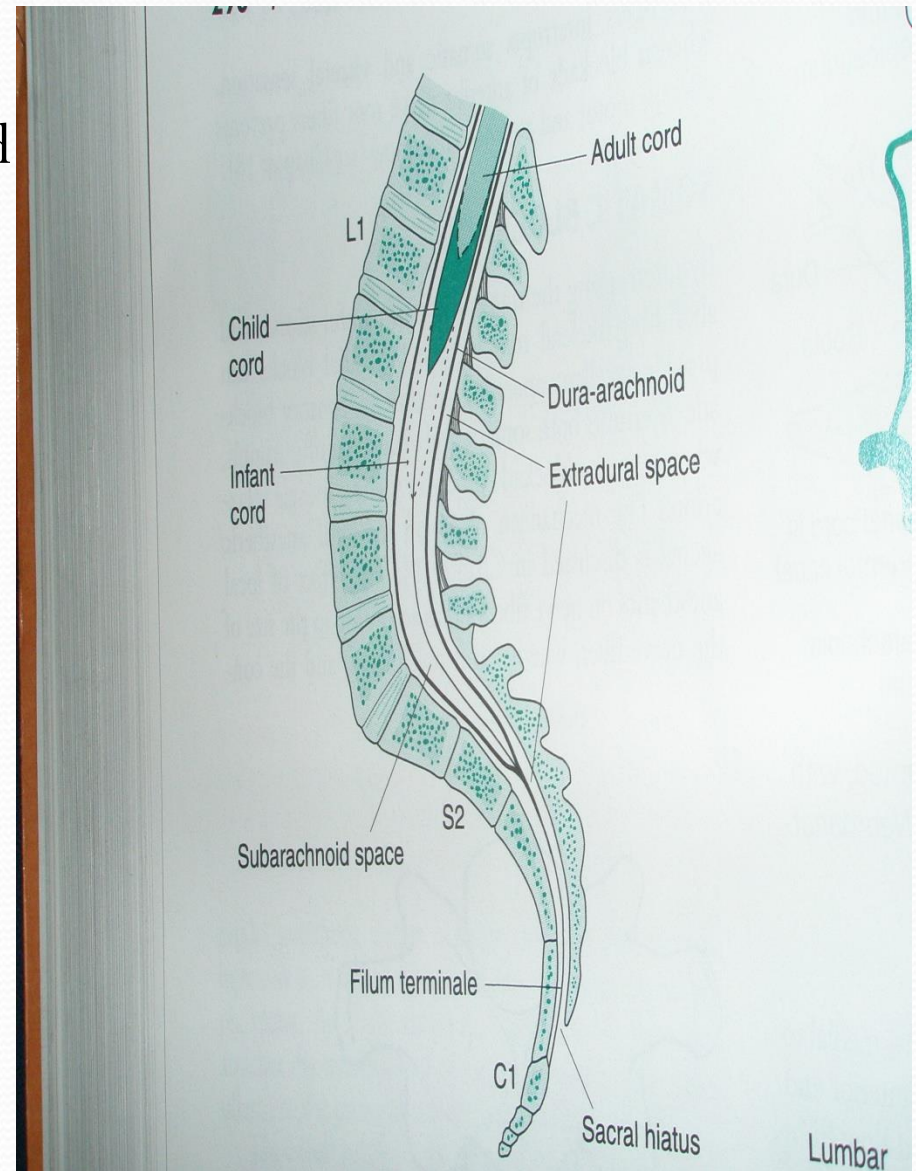
# Anatomy

- The caudal epidural space is the lowest portion of the epidural system and is entered through the **sacral hiatus**.
- The sacrum is a triangular bone that consists of the five fused sacral vertebrae (S1- S5).
- It articulates with the fifth lumbar vertebra and the coccyx.





- The sacral hiatus is an opening created by failure of the 5th sacral vertebra to fuse with others
- The sacral canal is a continuation of the lumbar spinal canal which terminates at the sacral hiatus.



# Choice of drugs & dosage

- Drugs that are commonly used include Lignocaine 1% and Bupivacaine 0.125 – 0.25%.



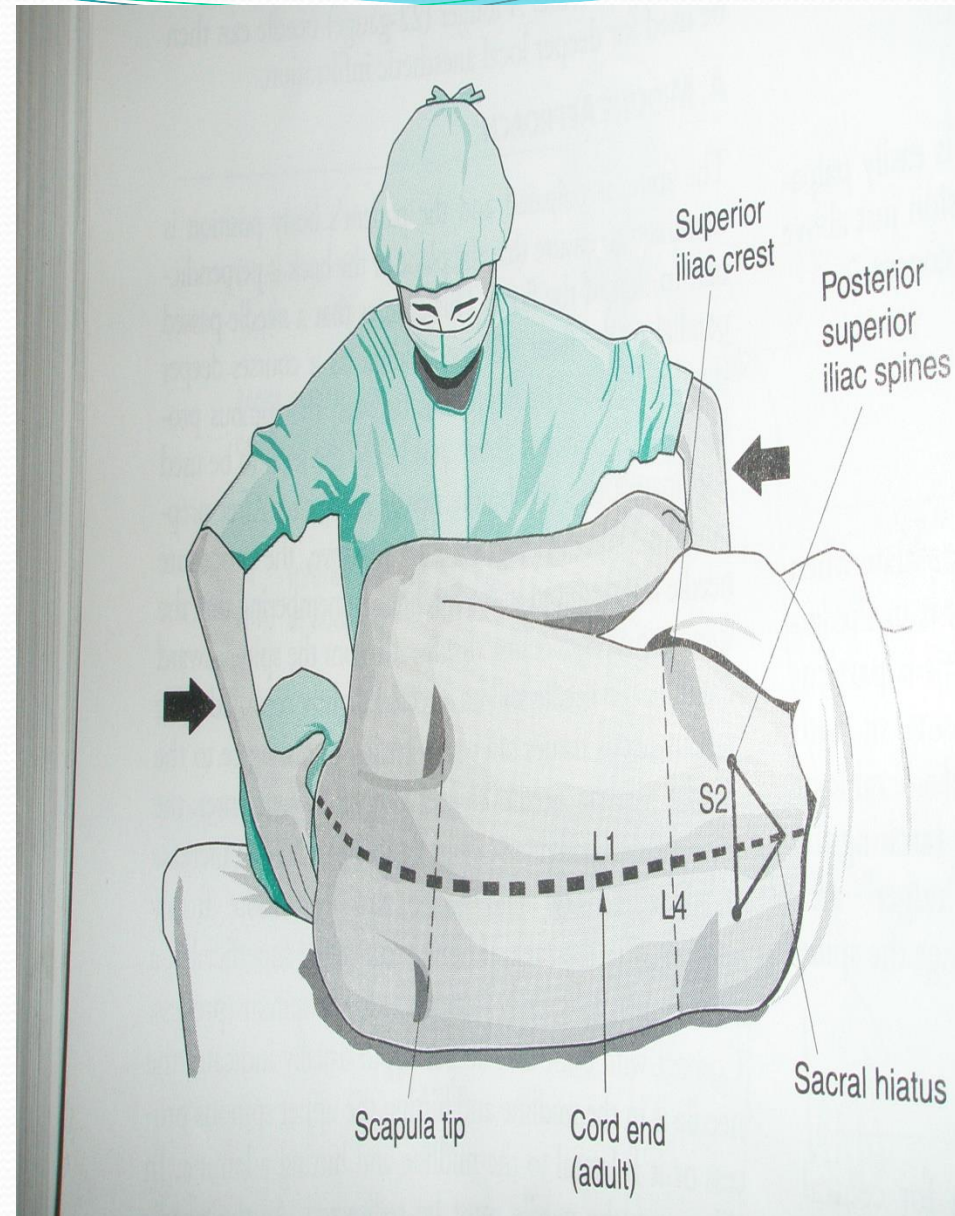
# Technique

- The patient is prepared as for general anesthesia:
  - All appropriate equipment for resuscitation must be available.
  - An iv cannula should always be inserted.
  - The procedure must be carried out with a strict aseptic technique.



- There are three main approaches:
  - the prone,
  - the semi-prone, and
  - the lateral.
- The later approach with fetus position is the most convenient and commonly used position
- caudal space is made more prominent by internally rotating the patient's ankles.

- The landmarks are palpated.
- The **sacral hiatus** and the **posterior superior iliac spines** form an equilateral triangle pointing inferiorly.
- The sacral hiatus can be located by first palpating the coccyx, and then sliding the palpating finger in a cephalad direction until a depression in the skin is felt.





Soliman, 2003

Iliac Crest

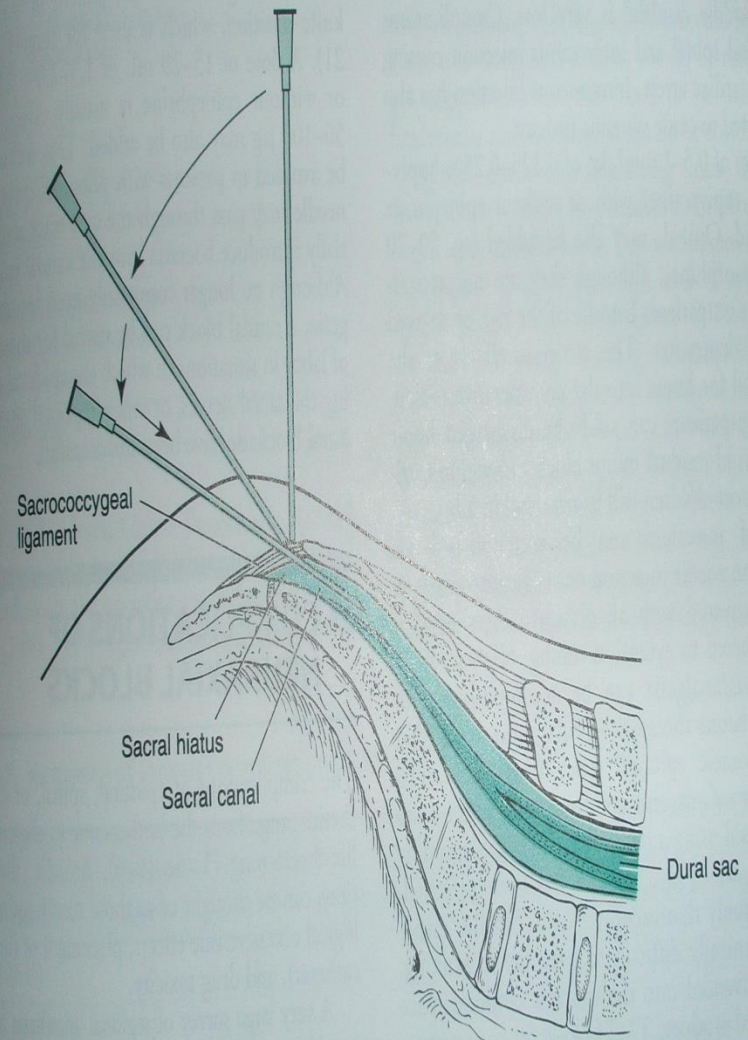
Coccyx

Sacral Hiatus

5th Sacral  
Cornua



- 7) Once the sacral hiatus is identified the area above is carefully cleaned with antiseptic solution, and
- a 22 gauge cannula or needle is directed at about 45-90 degree to skin and inserted till a "click" is felt as the sacro-coccygeal ligament is pierced.



**Figure 16-20.** Caudal block. Note the sacro-coccygeal ligament is penetrated with the needle almost at 90° and then must be angled down and advanced to enter the sacral hiatus.

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- Care should be taken not to insert the needle too far as the dura lies at or below the S2 level in the child.
  - The needle should be aspirated looking for either CSF or blood.
  - The injection should never be more than 10 ml/30 seconds



- Further tests to confirm the correct position include:
  - Introduction of a small amount of air will **not** produce subcutaneous emphysema, and will be heard as a "woosh" sound if a stethoscope is placed further up the lumbar spine.
  - There should be no resistance and local pain during injection.

- A small amount of local anesthetic should be injected as a test dose (2-4mls).
  - It should not produce either a lump in the subcutaneous tissues, or a feeling of resistance to the injection, nor any systemic effects such as arrhythmias or hypotension.
- If the test dose does not produce any side effects then the rest of the drug is injected, the needle removed and the patient positioned for surgery.



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- In the post-operative period, motor function must be checked and the patient should not be allowed to walk until complete return of motor function is assured.
  - The patient should not be discharged from hospital until he/she has passed urine, as urinary retention is a recognized complication.

# Complications

- Intravascular or intraosseous injection.
  - This may lead to seizure and/or cardio-respiratory arrest.
- Dural puncture
  - If this occurs then the patient will become rapidly apneic and profoundly hypotensive
- Perforation of the rectum
  - Contamination of the needle is extremely dangerous if it is then inserted into the epidural space.



# Combined spinal epidural anesthesia (CSEA)

# CSE anesthesia

- CSEA is an increasingly popular technique, with an estimated 10-fold increase in use in the past decade.
- Its advantages include:
  - rapid onset,
  - dense neuraxial block,
  - ability to titrate spread and duration of block, and
  - lower total drug dosage when compared to epidural anesthesia.



- Potential disadvantages include
  - increased time to perform the dual technique,
  - intrathecal migration of epidural drug and/ or catheter,
  - effects of increased epidural pressure from injection of solutions on spinal block, and decreased ability and reliability of epidural test.

# Patient Selection and Clinical Applications

- *Obstetrics*

- CSEA has been most widely accepted in the obstetric population.
- The concept of the “walking epidural” has become popular among patients, where intrathecal opioid allows rapid onset of analgesia without motor blockade.
- Lipid-soluble opioids, such as fentanyl (up to 25  $\mu\text{g}$ ) and sufentanil (up to 10  $\mu\text{g}$ ), are commonly used to provide dose-dependent analgesia for 60–90 minutes.



- Opioids are also commonly combined with small doses of local anesthetics, such as bupivacaine, to either **prolong** this initial spinal analgesia or to **reduce side effects** by decreasing the required dose of opioids.
- Previous dose response studies indicate that 2.5 mg of bupivacaine combined with either 15  $\mu\text{g}$  of fentanyl or 2.5  $\mu\text{g}$  of sufentanil provides satisfactory analgesia while reducing incidences of nausea and pruritus, when compared to larger doses of opioids.

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- As an anesthetic for cesarean section, CSEA offers a rapid, titratable block, good muscle relaxation, and the ability to use reduced doses of local anesthetic.
  - Several clinical trials have compared CSEA to epidural anesthesia with lidocaine or bupivacaine combined with fentanyl.
  - These studies report more rapid onset, better motor block, decreased anxiety levels, decreased shivering, and greater patient satisfaction with CSEA.



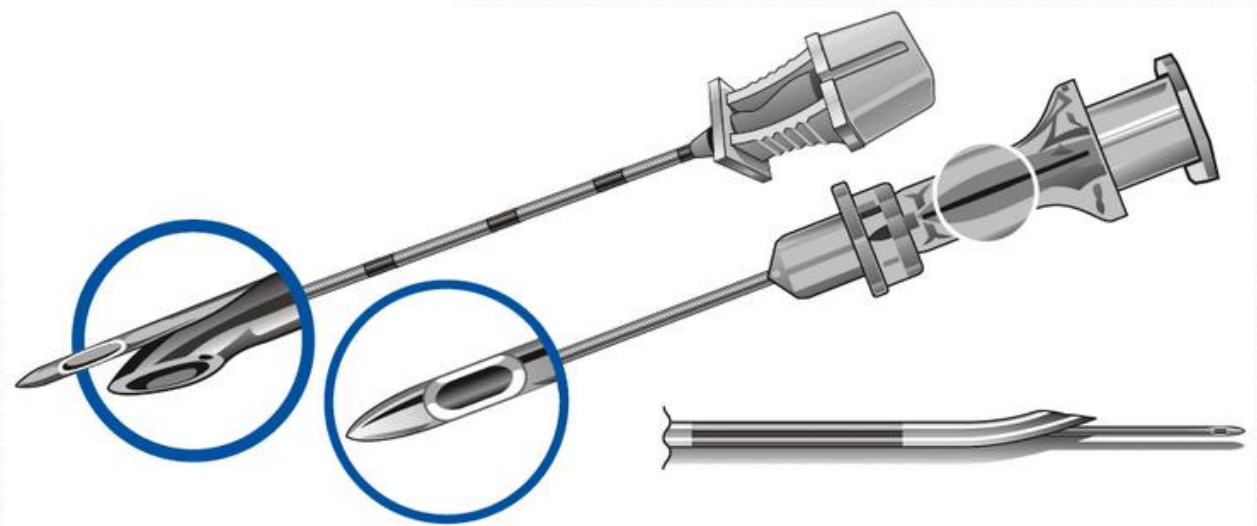
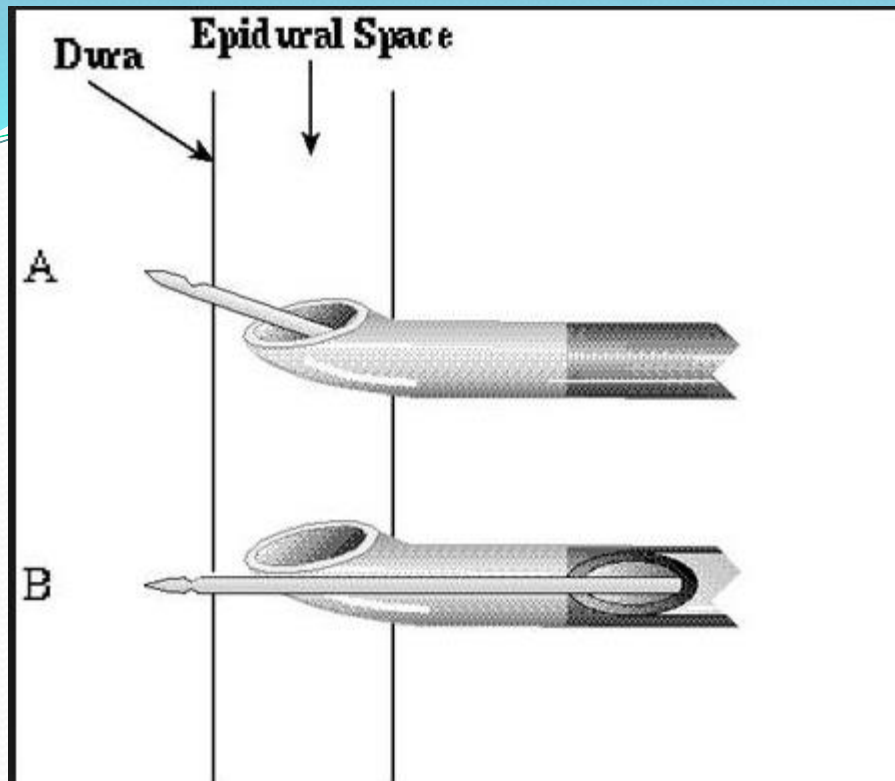
# ***Ambulatory Anesthesia***

- The dose of local anesthetic determines both anesthetic success and duration of recovery.
- Availability of the epidural catheter for a rescue anesthetic allows use of minimal doses of spinal local anesthetic with resultant rapid recovery and discharge and represents an alternative or complementary strategy to use of analgesic additives.

# Techniques and Equipment

- The needle-through needle technique is commonly used if such kit is available
  - A number of commercial kits are available.
  - The simplest version is a Tuohy needle (or equivalent) through which a long, small-gauge spinal needle (24–30 gauge) is passed.
  - Epidural needles with a “back hole” are also available, configured to allow placement of the spinal needle through a separate conduit so as to avoid angulation of the spinal needle





- Studies suggest that use of a back-hole needle may offer advantages over a conventional needle through needle technique.
  - A randomized trial in parturients observed decreased incidence of paresthesia (14% vs. 42%) and failure to obtain CSF on the first attempt (8% vs. 28%) with the back-hole needle.
- The separate conduit for the spinal needle may also reduce the theoretical risk of toxicity from metal fragments caused by needle friction.



- As an alternative to the needle through- needle technique, the double segment method also offers the ability to place the epidural catheter and administer a test dose prior to placing the spinal block.
  - Typically, the epidural and spinal portions are performed at different interspaces.
- By first introducing the catheter, there exists the potential risk of damaging the catheter with the spinal needle.
- Furthermore, creating two separate cutaneous punctures could lead to increased incidence of adverse events, including backache, headache, infection, and hematoma

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- A recent study demonstrated greater acceptance by surgical patients of the needle-through-needle over the double- segment technique (85% vs 67%).
  - That same study also showed a significantly longer time to perform the double-segment technique without decreasing the failure rate of spinal anesthesia, although other studies suggest a higher failure rate with the needle- through-needle technique.



# Potential Complications:

## *Failure of Spinal Anesthesia*

- The combined technique is associated with a higher failure rate of spinal anesthesia than conventional spinal anesthesia.
- There are a number of reasons for failure to occur.
  - (a) Smaller-gauge spinal needles with long lengths are typically used.
  - (b) Because the epidural needle has penetrated the tissue planes, there is little to anchor the spinal needle in place. Although a Luer lock apparatus is available, it locks at a fixed needle length and can result in not reaching or traversing the dura

# Spinal failure causes...

- (c) Any deviation from midline can lead to missing the dura altogether
- (d) If loss-of-resistance technique used saline, a false return of saline in the spinal needle rather than CSF can occur.
- (e) Finally, patient positioning and duration between spinal injection and completion of epidural catheter placement can change the characteristics of the spinal block.



# *Failure of Epidural Anesthesia*

- The incidence of failure is unlikely to be higher with the combined technique; however, the difficulty in early testing with a needle-through-needle technique may lead to late recognition of a misplaced catheter.
- Prior injection of spinal anesthetic precludes testing the epidural catheter for intra-thecal placement, and epidural injection of a test dose can lead to increased height of spinal block



- ***Intra-thecal Effects of Epidural Agent***

- The intra-thecal effects of epidurally administered drugs can occur through migration of the epidural catheter through the dural puncture, leakage of epidural anesthetic through the dural hole, and pressure effects of epidural injection.
- The likelihood of passing an epidural catheter through a dural hole is very small, provided a 24-gauge or smaller spinal needle is used



# Summary

- Combined spinal–epidural anesthesia (CSEA) is an increasingly popular technique
- studies report more rapid onset, better motor block, decreased anxiety levels, decreased shivering, and greater patient satisfaction with CSEA especially in obstetrics mothers.
- The combined technique is associated with a higher failure rate of spinal anesthesia than conventional spinal anesthesia.